Day 022 | Unit 02 | Teacher plan

Positive Rational Numbers: Fractions and Mixed Numbers

Check a Fraction Quotient

Standards: 6.NS.A.1

Learning target: I can check a Fraction Quotient, and explain and check my reasoning.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 2

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. If (a/b) ÷ (c/d) = q, then (c/d) × q = a/b. Multiplying by d/c undoes multiplying by c/d, so divide by a nonzero fraction by multiplying by its reciprocal. Simplify the product. Division and multiplication are inverse operations. For T ÷ s = n, the related equation is s × n = T. Label whether the unknown is the total, the group size, or the number of groups.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 15

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01 / Learn

If (a/b) ÷ (c/d) = q, then (c/d) × q = a/b. Multiplying by d/c undoes multiplying by c/d, so divide by a nonzero fraction by multiplying by its reciprocal. Simplify the product. Division and multiplication are inverse operations. For T ÷ s = n, the related equation is s × n = T. Label whether the unknown is the total, the group size, or the number of groups.

Worked example 1

Calculate 1/4 ÷ 1/4. Simplify.

1. Rename the numbers as fractions: 1/4 = 1/4 and 1/4 = 1/4.

2. Keep the first fraction and multiply by the reciprocal of the second: 1/4 × 4.

3. Multiply numerators and denominators: (1 × 4) / (4 × 1) = 4/4.

4. Simplify by dividing top and bottom by 4: 1.

5. Check the result: 1/4 × 1 = 1/4, so the quotient is correct.

Answer: 1

Worked example 2

The equation 1 × n = 3/2 represents equally sized portions. Find n, write the related division equation, and check.

1. Division and multiplication are inverse operations.

2. 3/2 ÷ 1 = 3/2 × 1 = 3/2.

3. Check: 1 × 3/2 = 3/2.

Answer: n = 3/2; 3/2 ÷ 1 = 3/2

Day 022 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Calculate 1 ÷ 1/2. Simplify.

02 / Constructed response

The equation 1/2 × n = 1 represents equally sized portions. Find n, write the related division equation, and check.

03 / Constructed response

Calculate 1/2 ÷ 1/4. Simplify.

04 / Constructed response

The equation 1/4 × n = 1/2 represents equally sized portions. Find n, write the related division equation, and check.

05 / Constructed response

Calculate 1/6 ÷ 5/4. Simplify.

06 / Constructed response

The equation 2/3 × n = 3/2 represents equally sized portions. Find n, write the related division equation, and check.

07 / Constructed response

Calculate 4/3 ÷ 1. Simplify and explain why your answer is reasonable.

08 / Model and constructed response

The equation 5/4 × n = 2 represents equally sized portions. Find n, write the related division equation, and check.

09 / Multiple choice

Calculate 3/2 ÷ 3/5. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.

A. 7/2
B. 5/2
C. 3/2
D. 2/5

10 / Constructed response

The equation 4/3 × n = 1/3 represents equally sized portions. Find n, write the related division equation, and check.

Day 022 | Unit 02 | Answer key

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1. 2

1 ÷ 1/2 = 1 × 2 = 2.

6.NS.A.1 | Constructed response

2. n = 2; 1 ÷ 1/2 = 2

1 ÷ 1/2 = 1 × 2 = 2. Check: 1/2 × 2 = 1.

6.NS.A.1 | Constructed response

3. 2

1/2 ÷ 1/4 = 1/2 × 4 = 2.

6.NS.A.1 | Constructed response

4. n = 2; 1/2 ÷ 1/4 = 2

1/2 ÷ 1/4 = 1/2 × 4 = 2. Check: 1/4 × 2 = 1/2.

6.NS.A.1 | Constructed response

5. 2/15

1/6 ÷ 5/4 = 1/6 × 4/5 = 2/15.

6.NS.A.1 | Constructed response

6. n = 9/4; 3/2 ÷ 2/3 = 9/4

3/2 ÷ 2/3 = 3/2 × 3/2 = 9/4. Check: 2/3 × 9/4 = 3/2.

6.NS.A.1 | Constructed response

7. 4/3

4/3 ÷ 1 = 4/3 × 1 = 4/3.

6.NS.A.1 | Constructed response

8. n = 8/5; 2 ÷ 5/4 = 8/5

2 ÷ 5/4 = 2 × 4/5 = 8/5. Check: 5/4 × 8/5 = 2.

6.NS.A.1 | Model and constructed response

9. B - 5/2

3/2 ÷ 3/5 = 3/2 × 5/3 = 5/2.

6.NS.A.1 | Multiple choice

10. n = 1/4; 1/3 ÷ 4/3 = 1/4

1/3 ÷ 4/3 = 1/3 × 3/4 = 1/4. Check: 4/3 × 1/4 = 1/3.

6.NS.A.1 | Constructed response